"Of all the implications of general relativity for the structure and evolution of the Universe, this question of the fate of great masses of matter is one of the most challenging," Wheeler asserted. On this his audience could agree. Wheeler then, in a near replay of Arthur Eddington's attack on Chandrasekhar twenty-four years earlier (Chapter 4), described Oppenheimer's view that massive stars must die by imploding to form black holes, and then he opposed it: Such implosion "does not give an acceptable answer," Wheeler asserted. Why not? For essentially the same reason as Eddington had rejected it; in Eddington's words, "there should be a law of Nature to prevent a star from behaving in this absurd way." But there was a deep difference between Eddington and Wheeler: Whereas Eddington's 1935 speculative mechanism to save the Universe from black holes was immediately branded as wrong by such experts as Niels Bohr, Wheeler's 1958 speculative mechanism could not at the time be proved or disproved—and fifteen years later it would turn out to be partially right (Chapter 12).

Wheeler's speculation was this. Since (in his view) implosion to a black hole must be rejected as physically implausible, "there seems no escape from the conclusion that the nucleons [neutrons and protons] at the center of an imploding star must necessarily dissolve away into radiation, and that this radiation must escape from the star fast enough to reduce its mass [below about 2 Suns]" and permit it to wind up in the neutron-star graveyard. Wheeler readily acknowledged that such a conversion of nucleons into escaping radiation was outside the bounds of the known laws of physics. However, such conversion might result from the as yet ill-understood "marriage" of the laws of general relativity with the laws of quantum mechanics (Chapters 12–14). This, to Wheeler, was the most enticing aspect of "the problem of great masses": The absurdity of implosion to form a black hole forced him to contemplate an entirely new physical process. (See Figure 6.1.)

Oppenheimer was not impressed. When Wheeler finished speaking, he was the first to take the floor. Maintaining a politeness that he had not displayed as a younger man, he affirmed his own view: "I do not know whether non-rotating masses much heavier than the sun really occur in the course of stellar evolution; but if they do, I believe their implosion can be described in the framework of general relativity [without asserting new laws of physics]. Would not the simplest assumption be that such masses